2008年工程经济学期末. 张志鹏

1. (1) 月实际利率 $i_1 = \frac{4\%}{12} = \frac{1}{3}\%$, $i_2 = \frac{6\%}{12} = \frac{1}{2}\%$

加息前月还款 $A_1 = 70万 \cdot (A/P, i_1, 180) = 5177.8元$

2009年末未还款 $P_0 = A_1 \cdot (P/A, i_1, 168) = 665231.4元$

加息后月还款 $A_2 = P_0 \cdot (A/P, i_2, 168) = 5862.3元$

$A_2 - A_1 = 5862.3 - 5177.8 = 684.5元$

(2) 提前还款额 $= I = P_0 + 5000 = 670231.4元$

设想一笔投资, 期初投入 I, 回本 A_2, 计算该投资的IRR (插值法)

$NPV(i_1) = A_2 \cdot (P/A, i_1, 168) - I = 8243元$

$NPV(i_2) = A_2 \cdot (P/A, i_2, 168) - I = -5000元$

于是 $IRR = i_1 + (i_2 - i_1) \cdot \frac{NPV(i_1)}{NPV(i_1) + |NPV(i_2)|} = 0.49\%$

亦即其他投资机会IRR低于0.49%时, 应提前还贷.

2. (1) $NPV = -(C_0)_0 + CI \cdot (P/A, 6\%, 5)$

计算五种情况有 $NPV_1 = 353.09$, $NPV_2 = 253.09$, $NPV_3 = 42.47$

$NPV_4 = -168.15$, $NPV_5 = -268.15$

对应发生概率 $P_1 = P_5 = 0.1$, $P_2 = P_4 = 0.2$, $P_3 = 0.4$

$E(NPV) = \sum P_i NPV_i = 42.48$ (万元)

$\sigma(NPV) = \sqrt{E(NPV^2) - [E(NPV)]^2} = 192.44$ (万元)

(2) $P_4 + P_5 = 30\%$

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